



Checking understanding of arithmetic sequences

Task A: Identifying arithmetic sequences

1) Label these sequences as 'arithmetic' or 'not arithmetic'.

a) 9, 13, 17, 21, 25, ...

$$\begin{array}{ccccccc} & \nearrow & & \nearrow & & \nearrow & \\ +4 & & +4 & & +4 & & +4 \end{array}$$

Arithmetic

b) 37, 31, 25, 19, 13, ...

$$\begin{array}{ccccccc} & \searrow & & \searrow & & \searrow & \\ -6 & & -6 & & -6 & & -6 \end{array}$$

Arithmetic

c) -70, -68, -66, -64, -62, ...

$$\begin{array}{ccccccc} & \nearrow & & \nearrow & & \nearrow & \\ +2 & & +2 & & +2 & & +2 \end{array}$$

Arithmetic

d) 35, 31, 26, 20, 13, ...

$$\begin{array}{ccccccc} & \searrow & & \searrow & & \searrow & \\ -4 & & -5 & & -6 & & -7 \end{array}$$

Not arithmetic

e) 12.6, 13.7, 14.8, 15.9 ...

$$\begin{array}{ccccccc} & \nearrow & & \nearrow & & \nearrow & \\ +1.1 & & +1.1 & & +1.1 & & \end{array}$$

Arithmetic

f) 3, 6, 12, 24, 48, ...

$$\begin{array}{ccccccc} & \nearrow & & \nearrow & & \nearrow & \\ +3 & & +6 & & +12 & & +24 \end{array}$$

Not arithmetic

g) $\frac{13}{29}, \frac{18}{29}, \frac{23}{29}, \frac{28}{29}, \dots$

$$\begin{array}{ccccccc} & \nearrow & & \nearrow & & \nearrow & \\ +\frac{5}{29} & & +\frac{5}{29} & & +\frac{5}{29} & & \end{array}$$

Arithmetic

h) $\frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \frac{1}{5}, \dots$

$$\begin{array}{ccccccc} & \searrow & & \searrow & & \searrow & \\ -\frac{1}{6} & & -\frac{1}{12} & & -\frac{1}{20} & & \end{array}$$

Not arithmetic

Task B: Checking understanding of arithmetic sequences

1) Find the first five terms of these three sequences:

a) $3n + 7$

Term number (n)	1	2	3	4	5
Term (T)	10	13	16	19	22

b) $3n - 7$

Term number (n)	1	2	3	4	5
Term (T)	-4	-1	2	5	8

c) $0.3n + 7$

Term number (n)	1	2	3	4	5
Term (T)	7.3	7.6	7.9	8.2	8.5

d) $7 - 3n$

Term number (n)	1	2	3	4	5
Term (T)	4	1	-2	-5	-8

2) Find the 10th, 50th, and 100th terms of these sequences.

a) $9n + 6$

$$n = 10, \quad T = 9n + 6 = 9(10) + 6 = 96$$

$$n = 50, \quad T = 9n + 6 = 9(50) + 6 = 456$$

$$n = 100, \quad T = 9n + 6 = 9(100) + 6 = 906$$

b) $9n - 6$

$$n = 10, \quad T = 9n - 6 = 9(10) - 6 = 84$$

$$n = 50, \quad T = 9n - 6 = 9(50) - 6 = 444$$

$$n = 100, \quad T = 9n - 6 = 9(100) - 6 = 894$$

c) $0.9n + 0.6$

$$n = 10, \quad T = 0.9n + 0.6 = 0.9(10) + 0.6 = 9.6$$

$$n = 50, \quad T = 0.9n + 0.6 = 0.9(50) + 0.6 = 45.6$$

$$n = 100, \quad T = 0.9n + 0.6 = 0.9(100) + 0.6 = 90.6$$

d) $6 - 9n$

$$n = 10, \quad T = 6 - 9n = 6 - 9(10) = -84$$

$$n = 50, \quad T = 6 - 9n = 6 - 9(50) = -444$$

$$n = 100, \quad T = 6 - 9n = 6 - 9(100) = -894$$

Task C: Representations of arithmetic sequences

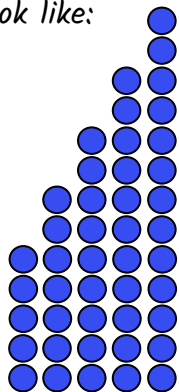
1) Draw a visual representation for the first five terms of these sequences:

a) $5, 7, 9, 11, 13, \dots$

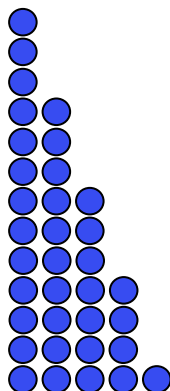
b) $13, 10, 7, 4, 1$

c) $5n + 1$

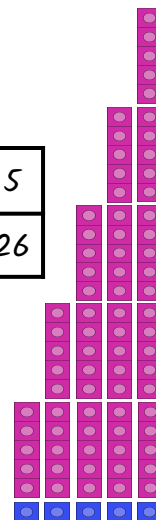
Examples of what they might look like:



In each case the linear structure of the sequence is revealed.



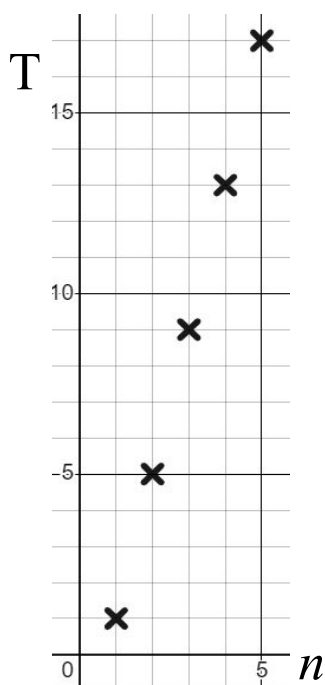
n	1	2	3	4	5
T	6	11	16	21	26



2) Plot the first five terms of these sequences:

a) $4n - 3$

n	1	2	3	4	5
T	1	5	9	13	17



b) $10 - 3n$

n	1	2	3	4	5
T	7	4	1	-2	-5

